

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	V Sanjana	Reg. No.:	192571005
Section / Batch:	2025	Date:	27-07-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I V Sanjana / 192571005 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

73. Linear search - search for repeated Non-consecutive Target

Given dataset :

[31, 14, 29, 14, 52, 67, 14]

Target element = 14

Problem Interpretation

The given dataset is unsorted. The value 14 occurs repeatedly at different nonconsecutive positions: index 1, index 3, and index 6. We must find only the first occurrence of 14.

Since the list is unsorted, Binary Search cannot be used. Linear search is suitable because it checks each element sequentially from the beginning until it finds the target.

The Search starts from index 0.

- At index 0, the element is 31. Since $31 \neq 14$, continue searching.
- At index 1, the element is 14. Since $14 = 14$, the first occurrence is found.

So, the first occurrence of 14 is at :

- index = 1
- Position = 2nd position
- Number of comparisons = 2

Verification

Dataset [31, 14, 29, 14, 52, 67, 14]

At index 1, the value is 14. Before this, index 0 contains 31, which is not equal to 14.

Therefore, the result is correct.

Sequential search process

Comparison	Index	Element checked	check	Result
1	0	31	$31=14?$	No
2	1	14	$14=14$	Yes.

Let n be the number of elements.

Case	Conditions	Comparison	Complexity
Best case	Target is first element	1	$O(1)$
Average case	Target is around the middle	$n/2$	$O(n)$
Worst case	Target is last or absent	n	$O(n)$

For this dataset, only 2 comparisons are needed because the first target value appears near the beginning.

Efficiency Interpretation:

Linear search is easy and useful for unsorted lists. It can find the first repeated target correctly because it always checks from left to right.

74. Binary Search - Target Near Upper Middle Region

Given Sorted dataset:

[2, 6, 10, 14, 18, 22, 26, 30, 34]

Target element = 22

Problem Interpretation

The given dataset is sorted in ascending order. Therefore, Binary Search is appropriate.

Binary search compares the target with the middle element. Based on the comparison, it removes either the left half or the right half of the list. This makes it much faster than Linear Search for large sorted datasets.

Initial values:

low = 0

high = 8

Formula:

$$\text{mid} = (\text{low} + \text{high}) // 2$$

Step 1

low = 0, high = 8

$$\text{mid} = (0 + 8) // 2 = 4$$

Element at index 4 is 18

Since:

$$22 > 18$$

The target must be in the right half.

New range:

low = 5, high = 8

Step 2

low = 5, high = 8

mid = $(5+8)//2 = 6$

Element at Index 6 is 26

Since:

$$22 < 26$$

The target must be in the left half.

New range:

low = 5, high = 5

Step 3

low = 5, high = 5

mid = $(5+5)//2 = 5$

Element at Index 5 is 22

Since:

$$22 = 22$$

The target is found

Result :

- The target element = 22
- Index = 5
- Position = 6th position
- Total comparisons = 3